Lab 6: Code and Branch Coverage Analysis

Student SRN: PES1UG24AM026 • 28 October 2026

Repository	SETAPESU26/L6_L7_26_loan_eligibility
Application under test	loan_eligibility.py
Code coverage test file	CodeCoverageTest_PES1UG24AM026_application.py
Branch coverage test file	BranchCoverageTest_PES1UG24AM026_application.py

1. Objective

This lab measures how much of the loan eligibility application is executed during automated testing and whether both outcomes of its decision branches are exercised.

2. Tools and environment

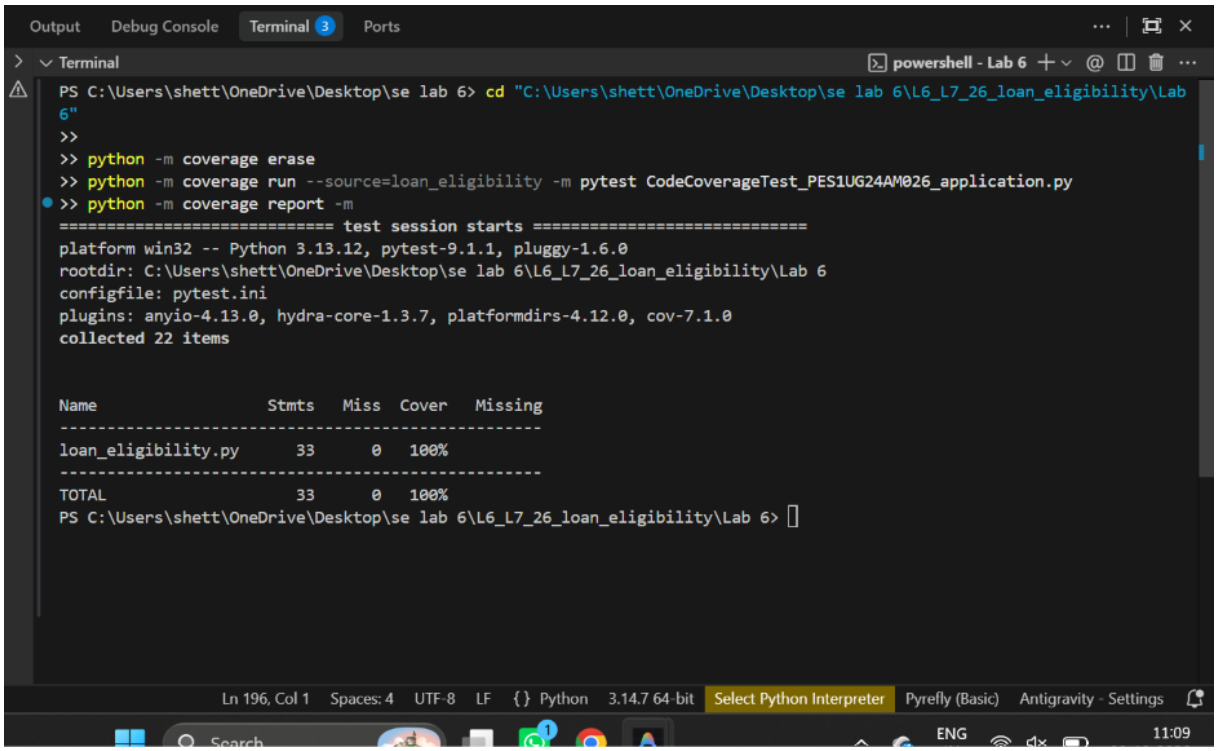
Tool	Version shown in terminal
Python	3.13.12
pytest	9.1.1
coverage.py	Version not visible in the supplied terminal screenshots

3. Code coverage analysis

The terminal output shows 22 test items collected and a coverage report for loan_eligibility.py with 33 statements, 0 missed statements, and 100% coverage. The tests cover credit-band mapping, eligibility conditions, boundary values, and interest-rate calculations.

```
Command used:
python -m coverage erase
python -m coverage run --source=loan_eligibility -m pytest CodeCoverageTest_PES1UG24AM026_application.py
python -m coverage report -m
```

Code coverage terminal output

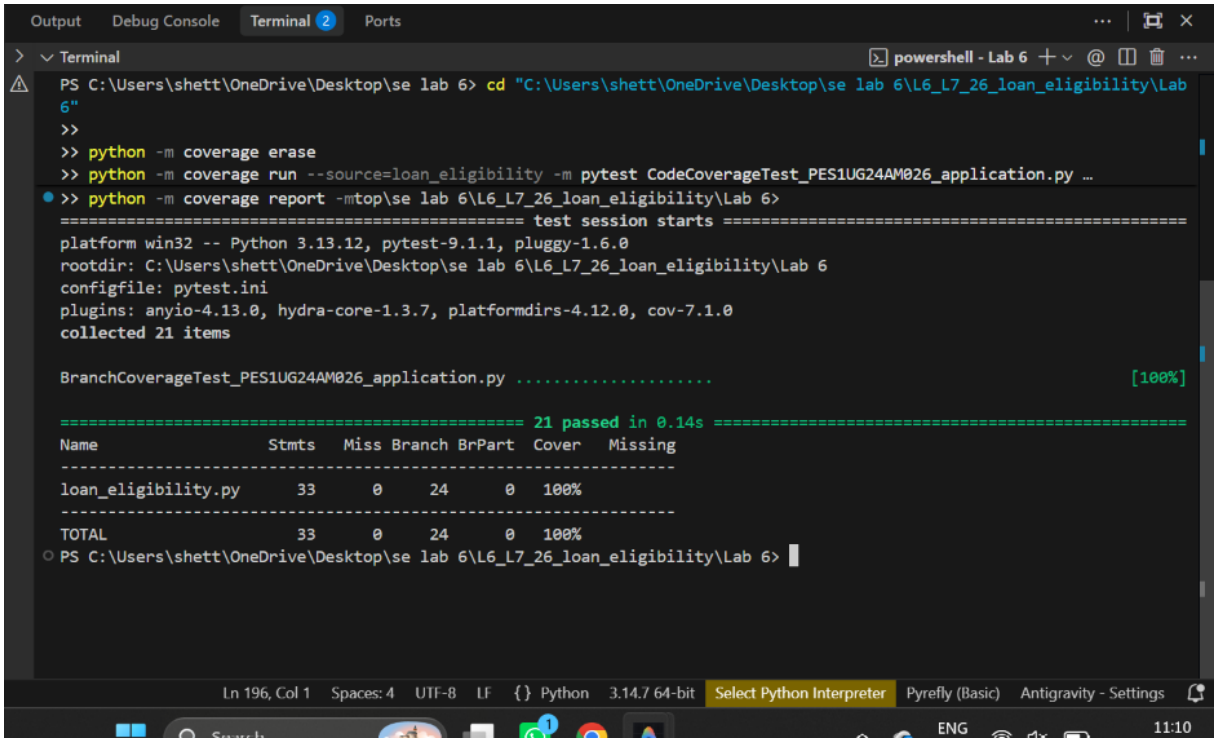

Output
Debug Console
Terminal 3
Ports
Terminal
powershell - Lab 6
PS C:\Users\shett\OneDrive\Desktop\se lab 6> cd "C:\Users\shett\OneDrive\Desktop\se lab 6\L6_L7_26_loan_eligibility\Lab 6"
>>
>> python -m coverage erase
>> python -m coverage run --source=loan_eligibility -m pytest CodeCoverageTest_PES1UG24AM026_application.py
>> python -m coverage report -m
============================= test session starts =============================
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\shett\OneDrive\Desktop\se lab 6\L6_L7_26_loan_eligibility\Lab 6
configfile: pytest.ini
plugins: anyio-4.13.0, hydra-core-1.3.7, platformdirs-4.12.0, cov-7.1.0
collected 22 items
Name Stmts Miss Cover Missing
loan_eligibility.py 33 0 100%
TOTAL 33 0 100%
PS C:\Users\shett\OneDrive\Desktop\se lab 6\L6_L7_26_loan_eligibility\Lab 6>
Ln 196, Col 1
Spaces: 4
UTF-8
LF
{} Python
3.14.7 64-bit
Select Python Interpreter
Pyrefly (Basic)
Antigravity - Settings
Search
ENG
11:09

4. Branch coverage analysis

The terminal output shows 21 tests passed in 0.14 seconds. The report for loan_eligibility.py lists 33 statements, 0 missed statements, 24 branches, 0 partially covered branches, and 100% coverage.

```
Command used:
python -m coverage erase
python -m coverage run --branch --source=loan_eligibility -m pytest
BranchCoverageTest_PES1UG24AM026_application.py
python -m coverage report -m
```

Branch coverage terminal output



5. Summary of results

Measure	Result	Requirement
Code coverage	100% (33 statements, 0 missed)	At least 90%
Branch coverage	100% (24 branches, 0 partial)	At least 90%
Code coverage tests	22 items collected	Target achieved
Branch coverage tests	21 passed	Target achieved

6. Conclusion

The recorded results meet the lab requirement of at least 90% code coverage and at least 90% branch coverage. The code coverage run reports 100% coverage, and the branch coverage run reports 100% coverage with no missed statements or partially covered branches.